

Jona Nakai
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Homework #5

$$1) p(y|\theta, n) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

$$p(\theta) = \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

$$p(y) = \int_0^1 p(y|\theta) p(\theta) d\theta = \int_0^1 \binom{n}{y} \theta^y (1-\theta)^{n-y} \cdot \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)} d\theta$$

$$= \frac{\binom{n}{y}}{B(\alpha, \beta)} \int_0^1 \theta^{y+\alpha-1} (1-\theta)^{n-y+\beta-1} d\theta = \frac{\binom{n}{y}}{B(\alpha, \beta)} B(y+\alpha, n-y+\beta)$$

$$p(\theta|y, n) = \frac{\left[\binom{n}{y} \theta^y (1-\theta)^{n-y} \right] \left[\frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)} \right]}{\frac{\binom{n}{y}}{B(\alpha, \beta)} B(y+\alpha, n-y+\beta-1)}$$

$$= \frac{\theta^{(y+\alpha)-1} (1-\theta)^{(n-y+\beta)-1}}{B(y+\alpha, n-y+\beta-1)} \Rightarrow \text{Beta}(y+\alpha, n-y+\beta)$$

Since the prior and posterior are both Beta distributions, the Beta is a conjugate prior for θ of the Binomial distribution

$$2. B(2, 2) = \frac{\Gamma(2) \Gamma(2)}{\Gamma(4)} = \frac{1! \cdot 1!}{3!} = \frac{1}{6}$$

$$p(x) = \frac{x^{2-1} (1-x)^{2-1}}{1/6} = 6x(1-x)$$

$$E(X) = \int_0^1 x^{2-1} 6x(1-x) dx = 6 \int_0^1 x^2 - x^3 dx = 6 \left(\frac{x^3}{3} - \frac{x^4}{4} \Big|_0^1 \right)$$

$$= 6 \left(\frac{1}{3} - \frac{1}{4} \right) = 6 \cdot \frac{1}{12} = \frac{1}{2} = 0.5$$

$$3a) p(y|\theta) = \theta e^{-\theta y}$$

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}$$

$$\begin{aligned} p(y) &= \int_0^\infty \theta e^{-\theta y} \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} d\theta \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \theta^\alpha e^{-(y+\beta)\theta} d\theta \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+1)}{(y+\beta)^{\alpha+1}} \end{aligned}$$

$$\begin{aligned} p(\theta|y) &= \theta e^{-\theta y} \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} \cdot \frac{\Gamma(\alpha)}{\beta^\alpha} \cdot \frac{(y+\beta)^{\alpha+1}}{\Gamma(\alpha+1)} \\ &= \frac{(y+\beta)^{\alpha+1}}{\Gamma(\alpha+1)} \theta^\alpha e^{-(y+\beta)\theta} \Rightarrow \text{Gamma}(\alpha+1, y+\beta) \end{aligned}$$

Prior and posterior are both gamma $\Rightarrow$ conjugate

$$b) p(y_i|\theta) = \theta^n e^{-\theta \sum_{i=1}^n y_i}$$

$$p(\theta) = \frac{1}{\Gamma(1)} \theta^{1-1} e^{-\theta} = e^{-\theta}$$

$$p(y) = \int_0^\infty \theta^n e^{-\theta(\sum_{i=1}^n y_i + 1)} d\theta = \frac{\Gamma(n+1)}{(\sum_{i=1}^n y_i + 1)^{n+1}}$$

$$p(\theta|y_i) = \theta^n e^{-\theta(\sum_{i=1}^n y_i + 1)} \frac{(\sum_{i=1}^n y_i + 1)^{n+1}}{\Gamma(n+1)} \Rightarrow \text{Gamma}(n+1, 1 + \sum_{i=1}^n y_i)$$

$$E(\theta|y_i) = \frac{n+1}{1 + \sum_{i=1}^n y_i}$$

$$c) p(y|\theta) = \frac{\theta^y e^{-\theta}}{y!} \quad p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}$$

$$p(y) = \int_0^\infty \frac{\theta^y e^{-\theta}}{y!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} d\theta$$

$$= \frac{\beta^\alpha}{y! \Gamma(\alpha)} \int_0^\infty \theta^{y+\alpha-1} e^{-\theta(1+\beta)} d\theta$$

$$= \frac{\beta^\alpha}{y! \Gamma(\alpha)} \cdot \frac{\Gamma(y+\alpha)}{(1+\beta)^{y+\alpha}}$$

$$p(\theta|y) = \frac{\theta^y e^{-\theta}}{y!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} \cdot \frac{y! \Gamma(\alpha)}{\beta^\alpha} \cdot \frac{(1+\beta)^{y+\alpha}}{\Gamma(y+\alpha)}$$

$$= \frac{\theta^{(y+\alpha)-1} e^{-\theta(1+\beta)}}{\Gamma(y+\alpha)} \cdot \frac{(1+\beta)^{y+\alpha}}{\Gamma(y+\alpha)}$$

$$\Rightarrow \text{Gamma}(y+\alpha, 1+\beta)$$

Prior and posterior are both gamma $\Rightarrow$ conjugate

d) From part (c), posterior $\Rightarrow \text{Gamma}(1 + \sum_{i=1}^n y_i, n+1)$

$$E(\theta|y_i) = \frac{1 + \sum_{i=1}^n y_i}{n+1}$$